GAMEATHON 2026

# MEMORY//LEAK

## Every defense leaves a pattern.

A 3D cybersecurity strategy game where an adaptive attacker learns from the player's defensive behaviour.

3D CYBERSECURITY • STRATEGY • ADAPTIVE ATTACKER

• C# • Blender • PC

TEAM THREATLOOP

01 / PROBLEM

# Cybersecurity awareness should be experienced, not only explained.

THE CURRENT GAP

- Presentations communicate warnings.
- Quizzes test memory.
- Real incidents require investigation and decisions.
- Fixed attack sequences become predictable.

OUR QUESTION

**What if cybersecurity awareness itself became the game?**

OUR APPROACH

**DETECT→INVESTIGATE→DEFEND→RECOVER→ADAPT**

**Cybersecurity is not a quiz layered onto a game.**

Player decisions directly affect attack progress, system integrity and mission success.

02 / GAME CONCEPT

# You defend NEXUS. ZERO is watching.

**NEXUS** Fictional organization whose users, accounts, endpoints, network and sensitive data are under attack.

**PLAYER** Cybersecurity analyst responsible for detecting, investigating and containing incidents.

**ZERO** Adaptive attacker that observes defensive behaviour and changes future attack strategies.



03 / CORE INNOVATION

# The attacker learns from how you defend.

ZERO TRACKS

- **Defense preference**
- Target priority
- Response pattern
- Neglected controls
- Response timing

**Repeated action = information for ZERO,
not automatically a mistake.**

ADAPTATION

| | | |
|---|---|---|
| 01 | OBSERVE | Record player behaviour. |
| 02 | PROFILE | Find patterns and neglected layers. |
| 03 | ADJUST | Change attack weights / strategy. |
| 04 | PRESSURE | Exploit the pattern in a future context. |

**RANDOMNESS decides WHAT may happen.**

**ZERO influences HOW intelligently it happens.**

04 / GAMEPLAY

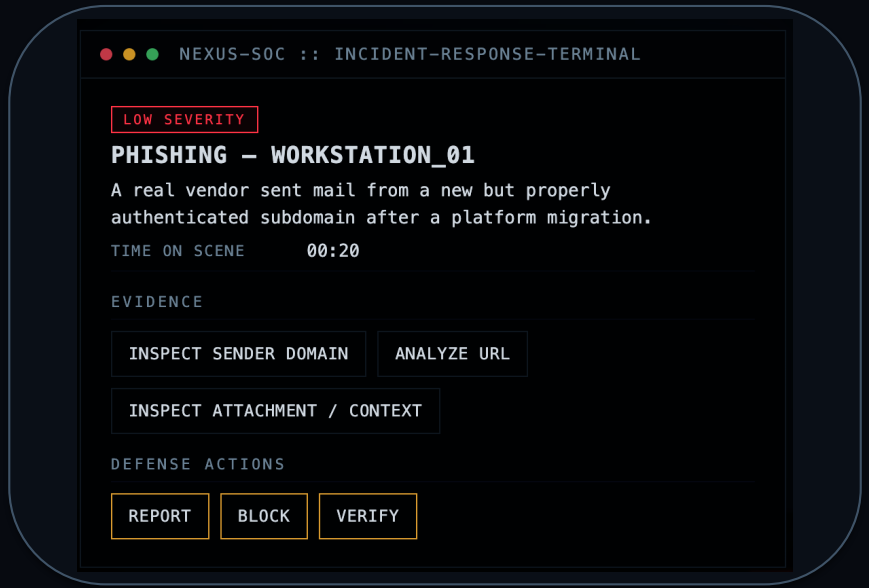
# One cyber incident becomes a playable decision loop.

ATTACK → ALERT → INVESTIGATE → DEFEND → CONSEQUENCE → RECOVER → ZERO ADAPTS

EXAMPLE: PHISHING

- Inspect sender, URL and message context.
- Identify malicious content.
- Choose the appropriate response.
- Experience the consequence.

SECURED • CONTAINED • BREACHED



05 / PROGRESSION

# 30 missions. 7 threat families. One personalized finale.

### THREAT FAMILIES

- PHISHING
- SOCIAL ENGINEERING
- CREDENTIAL THEFT
- MALWARE
- NETWORK INTRUSION
- DATA EXFILTRATION
- RANSOMWARE

### 6 LEVELS — 5 missions per level

| | |
|---|---|
| 1 | BEGINNER |
| 2 | EASY |
| 3 | MEDIUM |
| 4 | HARD |
| 5 | PRO |
| 6 | IMPOSSIBLE |

**IMPOSSIBLE Mission 5 :** personalized multi-stage ZERO finale

06 / PROTOTYPE

# A playable 3D browser prototype.




## IMPLEMENTATION

| | |
|---|---|
| 3D | Three.js |
| LOGIC | JavaScript |
| AUDIO | Web Audio API |
| SAVE | localStorage |
| WORLD | 3D SOC + workstations + server racks |
| SYSTEMS | Randomized threats + dynamic evidence + ZERO adaptation |

Core proof: investigation → defense → consequence → adaptation

# Players learn cybersecurity by making decisions.

**PHISHING**

Identify suspicious messages and links.

**IDENTITY**

Understand MFA, credentials and sessions.

**ENDPOINT**

Recognize malware and containment.

**INCIDENT RESPONSE**

Detect → Investigate → Contain → Recover.

**DEFENSE IN DEPTH**

Avoid over-reliance on one security layer.

**BEHAVIOURAL SECURITY**

Understand how predictability can be exploited.

**No quiz wall. The player experiences the threat, decision and consequence.**

07 / WHY IT STANDS OUT

# Cybersecurity isn't the lesson. Cybersecurity is the gameplay.

| | |
|---|---|
| CREATIVITY | Adaptive attacker learns from the player. |
| CYBERSECURITY | Real threats + investigation + defense. |
| EDUCATION | Evidence and consequences drive learning. |
| FEASIBILITY | Focused playable prototype + rule-based adaptation. |
| GAMEPLAY | Strategy + uncertainty + replayability. |

OUR DIFFERENCE

**Most cybersecurity games teach players what an attack is.**

**MEMORY//LEAK makes the attacker learn how the player defends.**

Every defense leaves a pattern.


MEMORY//LEAK
EVERY DEFENSE LEAVES A PATTERN.
TEAM THREATLOOP
GAMEATHON 2026
THANK YOU
FOR LISTENING, PLAVING,
AND BELIEVING IN A SAFER DIGITAL WORLD.
Cybersecurity isn't just a subject.
It's a shared responsibility.
NEXUS
PEOPLE | SECURITY | TOMORROW
DETECT
INVESTIGATE
DEFEND
RECOVER
ADAPT
THREAT CONTAINED
BUT THIS
IS NOT THE END.
ZERO IS WATCHING.
SAME ATTACKS.
DIFFERENT PLAYERS.
DIFFERENT OUTCOMES.
NEXUS
DEFEND
LEARN
IMPROVE
REPEAT
PLAY
LEARN
STAY AWARE
BUILD A
SAFER TOMORROW
BECAUSE
A MORE SECURE WORLD
STARTS WITH PEOPLE
A SAFER
TOMORROW
IS A BRIGHTER
TOMORROW.
MEMORY//LEAK • GAMEATHON 2026 • TEAM THREATLOOP
NEXUS // PEOPLE // SECURITY // TOMORROW